

Cloud Computing Lab 5 Report

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1. Assignment 1: AWS RDS PostgreSQL

The existing Drupal application running on Amazon EC2 was connected to an Amazon RDS PostgreSQL database. The RDS instance used PostgreSQL, database name **drupal**, port **5432**, and instance class **db.t4g.micro**. The RDS instance was configured without public internet access.

Resource	Configuration
EC2	Rohit-Drupal-Lab4, eu-north-1
RDS	rohit-drupal-lab5-pg
Engine	PostgreSQL
Database	drupal
Port	5432
Drupal	11.4.7

RDS Connectivity and CRUD

Drupal successfully reported **Database: Connected** and **Drupal bootstrap: Successful** while using the RDS PostgreSQL endpoint. CRUD was demonstrated using Drupal nodes: a node was created, read, updated, deleted, and then verified as deleted. The final test created Node 9, read its published status, updated its title, deleted it, and confirmed that it no longer existed.

RDS security was configured so PostgreSQL access was restricted to the EC2 security group. The EC2 security group also contained HTTP port 80 for the running Drupal application.

2. Assignment 2: Amazon DynamoDB

A DynamoDB table named **Lab5DrupalNoSQL** was configured in eu-north-1 using on-demand capacity. The table uses **id** as its partition key. The implementation demonstrated the required DynamoDB data types: String (S), Number (N), Boolean (BOOL), List (L), and Map (M).

DynamoDB CRUD and IAM Security

Create, Read, Update, and Delete operations were demonstrated for the DynamoDB data. Access was configured through the EC2 IAM role **DrupalLab5SecretsRole** rather than hardcoded AWS access keys. Evidence was captured for the IAM policy and the EC2 instance role.

3. Application and Evidence

The Drupal application was verified from the EC2 instance with an HTTP 200 response. Application URL used during the lab: **http://13.60.27.33**.

The GitHub repository contains the README and the uploaded Lab 5 evidence screenshots. The screenshots cover the EC2 application, RDS connection, CRUD operations, DynamoDB data types and CRUD, IAM role/policy, and EC2 security-group configuration.

4. Conclusion

Lab 5 demonstrated integration of an EC2-hosted Drupal application with managed AWS database services. PostgreSQL data was migrated and accessed through RDS, CRUD operations were verified, DynamoDB CRUD and required data types were demonstrated, and IAM-based access was configured for DynamoDB. Security-group restrictions were applied to control database access.

GitHub repository: github.com/RohitRaina014-afk/cloud-computing-lab5